

Belief—MCTS—STREET—CRF: A Hybrid, Sample-Efficient Solver for Multi-Player No-Limit Texas Hold'em

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Abstract

We present Belief—MCTS—STREET—CRF, a multi-player No-Limit Hold'em agent coupling a transformer-style public encoder with a street-structured CRF (STREET—CRF) for opponent modeling, belief-aware MCTS with risk-sensitive selection, and lightweight endgame subgame re-solving via CFR^+ . Our core contributions are: (i) Consistency—Projected Beliefs (CPB), a maximum-entropy projection onto a group-invariant, realizable family used as root chance; (ii) Leakage-Free Public-Subgame CFR^+ with a symmetry-invariance guarantee under CPB; and (iii) IS—RS—CVaR—UCT, combining information-set root sampling with a streaming RU CVaR estimator and bandit-gated double progressive widening (DPW) for continuous sizes. We emphasize innovations that improve algorithmic efficiency and mathematical rigor: strong convexity and conditioning bounds with a reference-metric view for CPB; selection stability bounds for CVaR—UCT; and almost-sure densification guarantees for DPW. All reported numbers are deterministically generated by the embedded `simulation.py` and are reproduced byte-exactly in the included CSVs and `results.txt`.

1 Introduction

No-Limit Texas Hold'em (NLHE) is a canonical imperfect-information benchmark combining hidden private cards, public community cards, and large continuous action spaces. Milestone systems include DeepStack [1], safe/nested subgame solving and DCFR [2, 3, 4, 5], Pluribus [6], and public-belief search (ReBeL) [7]. Our objective is a practical, sample-efficient hybrid suitable for 6-max with modest compute.

Contributions (innovation, efficiency, and rigor): - STREET—CRF opponent model producing calibrated posteriors with a single trajectory-wide base measure and consistent bucketization. - IS—RS—CVaR—UCT: belief-aware MCTS with a streaming RU CVaR estimator and bandit-gated DPW for continuous sizes; we provide selection stability bounds. - Leakage-Free Public-Subgame CFR^+ using CPB as root chance with symmetry-invariance. - Analysis of a preconditioned, accelerated CPB solver with condition-number bounds, a reference-metric whitening perspective, and a continuation scheme; diagnostics use a simple GD baseline for transparency. - Reproducible, deterministic diagnostics: CPB projection residuals, IS—RS unbiasedness with CRN, RU—CVaR convergence on Uniform $[-1, 1]$, and DPW densification.

2 Related Work

DeepStack [1]; safe/nested subgame solving [2]; DCFR [3]; CFR [4]; CFR^+ [5]; Pluribus [6]; ReBeL [7]; POMCP [8]; ISMCTS [9]; DPW [10]; CVaR and optimization [11, 12, 13]; calibration [14]; PSRO [15]; AIVAT [16]; SMC [17]; stochastic approximation [18]; I-projections [19]; exchangeability [20]; UCT [21]; iterative scaling and maxent RF [22, 23]; mirror descent [24]; LinUCB [25]; information geometry and exponential families [29, 30]; accelerated first-order methods [27]; adaptive restarts [36]; Barzilai–Borwein steps [31]; L-BFGS [32]; DKW concentration [33]; probabilistic foundations [34]; risk-constrained RL with CVaR [37].

3 Preliminaries

We consider an N -player EFG with imperfect information $(\mathcal{S}, \mathcal{A}, \mathcal{N}, \mathcal{I}, \mathcal{C}, u)$. A public state x is a projection of a history h . Let H_{-i} be opponents' private hands and H_i the acting player's hand. A public belief state is $\rho_t(H_{-i} \mid h_t, H_i)$; feasibility masks enforce disjointness and blockers.

4 Public subgames and admissible chance

We re-solve public subgames $\mathcal{G}_{\text{pub}}(x, H_i)$ whose root chance samples opponents' private cards from $q(H_{-i} \mid x, H_i)$ supported on feasible hands and excluding H_i . A symmetry subgroup $G(x)$ permutes indistinguishable opponent labels. We require q to be $G(x)$ -invariant and Borel-measurable in (x, H_i) .

5 Method

5.1 Public encoder

A transformer encoder outputs masked policy $\pi_\theta(\cdot \mid x)$ and value $v_\theta(x)$ using tokenized public histories with position/card embeddings.

5.2 Street-CRF opponent modeling

We model streetwise latent styles and action features, inducing bucketed likelihoods $p_\phi(b \mid C(H_j), x)$ under a single base measure per trajectory (PMF over buckets in pot-fraction). Buckets are left-closed, right-open (last bucket closed); historical sizes are mapped deterministically across street boundaries by applying the same pot-fraction partition and boundary convention to past sizes. This order-preserving mapping ensures consistent likelihoods.

Representative mapping r_x uses bucket midpoints in pot-fraction, converted to chips at execution. Under a piecewise-Lipschitz value function with finitely many thresholds per state, midpoint representatives incur bounded error.

Lemma 1 (Midpoint representative error). *Let $I = [a, b]$ be a bucket interval in pot-fraction and suppose the normalized value V is L -Lipschitz on I . Then choosing the midpoint $a^* = (a + b)/2$ yields $|V(a^*) - \mathbb{E}_{U \sim \text{Unif}(I)}[V(U)]| \leq L(b - a)/4$ and $|V(a^*) - V(u)| \leq L(b - a)/2$ for any $u \in I$.*

Proof. The worst-case pointwise deviation from any $u \in I$ is bounded by $L|a^* - u| \leq L(b - a)/2$. The mean deviation bound follows by Jensen and symmetry. $\square$

Joint belief update and folds. With feasibility mask $\mathcal{H}(x, H_i)$, for opponent j acting with observed action a_t mapped to bucket b_t ,

$$\rho_{t+1}(H_{-i} \mid h_{t+1}, H_i) \propto \mathbf{1}\{H_{-i} \in \mathcal{H}(x_t, H_i)\} \rho_t(H_{-i} \mid h_t, H_i) p_\phi(b_t \mid C(H_j), x_t).$$

Folds contribute a single factor $p_\phi(\text{fold} \mid C(H_j), x_t)$; thereafter seat j is marked inactive for future action likelihoods, but H_j remains in the joint assignment for blockers and showdowns.

Proposition 1 (Posterior factorization under inactivity). *Let $\mathcal{A}_t$ be the set of active opponents at time t . Suppose $j \notin \mathcal{A}_{t'}$ for all $t' > t$. Then posterior updates after t include no further likelihood terms for H_j beyond feasibility, i.e., folding induces a single likelihood factor at t and no additional posterior drift.*

5.3 Consistency-Projected Beliefs (CPB)

We project an empirical posterior onto a realizable, $G(x)$ -invariant exponential family relative to a symmetric base measure using an I-projection with moment constraints [19], realized on an orbit-completed finite

support. Let the standardized statistics on orbit representatives be $S_o \in \mathbb{R}^K$ and define the exponential-family law on the finite support by $q_\lambda(o) \propto \exp(\langle \lambda, S_o \rangle)$, $Z(\lambda) = \sum_o \exp(\langle \lambda, S_o \rangle)$. We solve the penalized dual

$$\min_{\lambda \in \mathbb{R}^K} f_\gamma(\lambda) := \log Z(\lambda) - \langle \lambda, \hat{s} \rangle + \frac{\gamma}{2} \|\lambda\|_2^2,$$

which yields a unique optimizer for any $\gamma > 0$.

Theorem 1 (Strong convexity, Lipschitz gradient, condition number). *Let $\{S_o\}_{o=1}^m \subset \mathbb{R}^K$ be standardized orbit representatives, $R = \max_o \|S_o\|_2 < \infty$. For any $\gamma > 0$, f_γ is γ -strongly convex and has L -Lipschitz gradient with $L \leq R^2 + \gamma$. The condition number satisfies $\kappa \leq (R^2 + \gamma)/\gamma$. Consequently, fixed-step gradient descent with $\eta \in (0, 1/L]$ converges linearly to λ_γ^* .*

Proof. $\nabla f_\gamma(\lambda) = \mathbb{E}_{q_\lambda}[S] - \hat{s} + \gamma\lambda$, $\nabla^2 f_\gamma(\lambda) = \text{Cov}_{q_\lambda}(S) + \gamma I \succeq \gamma I$. Since $\lambda_{\max}(\text{Cov}(S)) \leq \mathbb{E}\|S\|^2 \leq R^2$, $L \leq R^2 + \gamma$. $\square$

Corollary 1 (Local curvature at the optimizer). *Let λ_γ^* minimize f_γ . Then the local gradient Lipschitz constant satisfies $L_* = \lambda_{\max}(\text{Cov}_{q_{\lambda_\gamma^*}}(S)) + \gamma \leq R^2 + \gamma$. A practical fixed step $\eta \approx 1/L_*$ is admissible and typically larger than $1/(R^2 + \gamma)$ due to standardization.*

Proposition 2 (Accelerated, preconditioned complexity). *Let $\mu = \gamma$ and $L \leq R^2 + \gamma$ as in Theorem 1. An accelerated gradient method with diagonal preconditioning and restarts attains $f_\gamma(\lambda^{(t)}) - f_\gamma(\lambda^*) \leq \mathcal{O}\left((1 - \sqrt{\mu/L})^t\right)$ and ε -accuracy in $\mathcal{O}\left(\sqrt{L/\mu} \log(1/\varepsilon)\right)$ gradients [27, 36]. Each iteration costs $\Theta(Km)$ with two passes (scores, expectation) using a stable log-sum-exp.*

Lemma 2 (Reference-metric whitening and condition number). *Let $\Sigma_{\text{ref}} = \frac{1}{m} \sum_{o=1}^m S_o S_o^\top$ and define whitened statistics $\tilde{S}_o = \Sigma_{\text{ref}}^{-1/2} S_o$, reparameterizing with $\tilde{\lambda} = \Sigma_{\text{ref}}^{1/2} \lambda$. Then for the penalized dual in these coordinates, $\nabla^2 \tilde{f}_\gamma(\tilde{\lambda}) = \text{Cov}_{q_{\tilde{\lambda}}}(\tilde{S}) + \gamma I$, so $L_{\text{ref}} \leq \lambda_{\max}(\mathbb{E}_{q_{\tilde{\lambda}}}[\tilde{S} \tilde{S}^\top]) + \gamma$. Along the uniform reference, $L_{\text{ref}} \leq 1 + \gamma$. Uniformly over all q , $L \leq R^2 + \gamma$.*

Proof. Change of variables yields the displayed Hessian. For the uniform reference, $\mathbb{E}[\tilde{S} \tilde{S}^\top] = I$. Taking spectral maxima gives the bounds. $\square$

Lemma 3 (Penalization bias under local strong convexity). *Assume the unpenalized dual f_0 has a unique minimizer λ_0^* and is μ_0 -strongly convex in a neighborhood of λ_0^* (e.g., $\lambda_{\min}(\text{Cov}_{q_{\lambda_0^*}}(S)) \geq \mu_0 > 0$). Then for $\gamma \in (0, \mu_0)$, $\|\lambda_\gamma^* - \lambda_0^*\|_2 \leq \frac{\gamma}{\mu_0 - \gamma} \|\lambda_0^*\|_2$, and the induced moment bias obeys $\|\mathbb{E}_{q_{\lambda_\gamma^*}}[S] - \mathbb{E}_{q_{\lambda_0^*}}[S]\|_2 \leq (R^2 + \gamma) \|\lambda_\gamma^* - \lambda_0^*\|_2$.*

Proof. Optimality gives $\nabla f_0(\lambda_\gamma^*) = -\gamma \lambda_\gamma^*$. Local strong convexity yields $\|\lambda_\gamma^* - \lambda_0^*\| \leq \frac{1}{\mu_0} \|\nabla f_0(\lambda_\gamma^*) - \nabla f_0(\lambda_0^*)\| = \frac{\gamma}{\mu_0} \|\lambda_\gamma^*\|$. Triangle inequality and rearrangement give the first bound. The moment bound follows from the mean value theorem and the Jacobian norm bound $\leq R^2 + \gamma$. $\square$

Proposition 3 (Geometric continuation and target bias). *Let $\gamma_t = \gamma_0 \rho^t$ with $\rho \in (0, 1)$. Warm-start stage t at $\lambda_{\gamma_{t-1}}^*$ and run an accelerated method with rate $(1 - \sqrt{\gamma_t/L_t})$ (with $L_t \leq R^2 + \gamma_t$). Then the total gradient evaluations $\sum_{t=1}^T \mathcal{O}\left(\sqrt{L_t/\gamma_t} \log \frac{C}{\varepsilon}\right) = \mathcal{O}\left(\sqrt{1 + \frac{R^2}{\gamma_T}} \log \frac{C}{\varepsilon} \cdot T\right)$, where C absorbs warm-start error via Lemma 3. For a target moment bias $\delta > 0$, a rule-of-thumb chooses γ_T by solving $(R^2 + \gamma_T) \frac{\gamma_T}{\mu_0 - \gamma_T} \|\lambda_0^*\| \leq \delta$ or, conservatively, $\gamma_T \approx \delta \mu_0 / (R^2 \|\lambda_0^*\|)$ when $\gamma_T \ll \mu_0$.*

Sketch. Sum the stage complexities and apply Lemma 3. The bias rule follows by imposing the moment-bias bound. $\square$

Preconditioner and stopping. We use a diagonal preconditioner $D_t = \text{diag}((v_t + \epsilon)^{-1/2})$ with $v_t = \rho v_{t-1} + (1 - \rho) g_t \odot g_t$, $\rho \in [0.9, 0.99]$, $\epsilon > 0$; this approximates Fisher/covariance scaling and is stable under bounded curvature of the softmax log-partition. Recommended stopping criteria: $\|\nabla f_\gamma(\lambda)\|_2 \leq 10^{-6}$ and moment residual $\|\mathbb{E}_{q_\lambda}[S] - \hat{s}\|_2 \leq 10^{-3}$.

Algorithm 1 Preconditioned–Accelerated CPB (single γ stage)

Require: Standardized stats $\{S_o\}$, target $\hat{s}$, penalty $\gamma > 0$, tol ε , radius R , EMA ρ , jitter ϵ

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1: Initialize  $\lambda^{(0)} = 0$ ,  $y^{(0)} = \lambda^{(0)}$ ,  $\theta^{(0)} = 1$ ,  $v^{(0)} = \mathbf{0}$ ; stepsize  $\eta \leftarrow 1/(R^2 + \gamma)$ 
2: for  $t = 0, 1, 2, \dots$  do
3:   Compute scores  $u_o = \langle y^{(t)}, S_o \rangle$ , weights  $w_o \propto \exp(u_o - \max_o u_o)$ ,  $Z = \sum_o w_o$ 
4:    $\mathbb{E}[S] \leftarrow \sum_o \frac{w_o}{Z} S_o$ ,  $g^{(t)} \leftarrow \mathbb{E}[S] - \hat{s} + \gamma y^{(t)}$ 
5:    $v^{(t+1)} \leftarrow \rho v^{(t)} + (1 - \rho) g^{(t)} \odot g^{(t)}$ ,  $D^{(t+1)} \leftarrow \text{diag}((v^{(t+1)} + \epsilon)^{-1/2})$ 
6:    $\lambda^{(t+1)} \leftarrow y^{(t)} - \eta D^{(t+1)} g^{(t)}$ 
7:    $\theta^{(t+1)} \leftarrow \frac{1 + \sqrt{1 + 4(\theta^{(t)})^2}}{2}$ ;  $y^{(t+1)} \leftarrow \lambda^{(t+1)} + \frac{\theta^{(t)} - 1}{\theta^{(t+1)}} (\lambda^{(t+1)} - \lambda^{(t)})$ 
8:   if  $\|\nabla f_\gamma(\lambda^{(t+1)})\|_2 \leq \varepsilon$  and  $\|\mathbb{E}_{q_{\lambda^{(t+1)}}}[S] - \hat{s}\|_2 \leq 10^{-3}$  then break
9:   end if
10:  Optional restart if  $f_\gamma(\lambda^{(t+1)}) > f_\gamma(\lambda^{(t)})$ 
11: end for
12: return  $\lambda^* \leftarrow \lambda^{(t+1)}$ 
```

5.4 Leakage–Free Public–Subgame CFR⁺

We re–solve with root chance $q = \tilde{\rho}_t(\cdot \mid x, H_i)$, equivariant menu generation, label–agnostic seeds, and permutation–equivariant regret–matching+. This yields $G(x)$ –invariant strategies at every iteration.

Definition 1 (Group action on information sets). *Let $G(x)$ act on player labels and induce a bijection $g : \mathcal{I} \rightarrow \mathcal{I}$, preserving public states and legal actions, with pullbacks on strategies and regret vectors via $(g \cdot \pi)(I, a) = \pi(g^{-1}I, g^{-1}a)$ and $(g \cdot r)(I, a) = r(g^{-1}I, g^{-1}a)$.*

Theorem 2 (Symmetry preservation of LF–PS–CFR⁺). *Suppose (i) the root chance q is $G(x)$ –invariant; (ii) action menus are $G(x)$ –equivariant; and (iii) regret matching+ is $G(x)$ –equivariant (e.g., orbit–averaged features and label–agnostic seeds). Then all CFR⁺ iterates are $G(x)$ –invariant.*

Proof. Under (i)&(ii) counterfactual values satisfy $v(I, a) = v(gI, ga)$. Regret accumulation and the regret–to–policy map commute with g under (iii), so if π^t is invariant then so is π^{t+1} . The base iterate is invariant by construction, hence all iterates are. $\square$

5.5 Belief–MCTS with CVaR–UCT and DPW

We use IS–RS with a frozen root CPB. Opponent actions in simulations are sampled from a known H –independent generator $\pi_{\text{opp}}(b \mid x) = (1 - \eta)A_x(\pi_\theta)(b \mid x) + \eta p_\phi^{\text{marg}}(b \mid x)$, renormalized over legal buckets. Public reveals are H –independent conditioned on feasibility.

Normalization and selection stability. Let $X \in [-1, 1]$ be normalized returns and $Y = -X$ losses.

The selection score is $U(x, a) = \hat{\mu}_{x,a} - \lambda \widehat{\text{CVaR}}_\alpha(Y) + c_{\text{puct}} \tilde{P}(a \mid x) \frac{\sqrt{\sum_b N(x, b)}}{1 + N(x, a)}$. With $|\hat{\mu}_{x,a}| \leq 1$ and $|\widehat{\text{CVaR}}_\alpha(Y)| \leq 1$, we have $|\hat{\mu}_{x,a} - \lambda \widehat{\text{CVaR}}_\alpha(Y)| \leq 1 + \lambda$, supporting co–scaling $c_{\text{puct}} \approx 1 + \lambda$.

Proposition 4 (Score sensitivity and ordering stability). *Suppose per–arm estimation errors satisfy $|\hat{\mu}_{x,a} - \mu_{x,a}| \leq \epsilon_\mu$ and $|\widehat{\text{CVaR}}_\alpha(Y_{x,a}) - \text{CVaR}_\alpha(Y_{x,a})| \leq \epsilon_c$, and exploration terms are identical across compared actions. If two actions a, a' satisfy $|(\mu_{x,a} - \lambda \text{CVaR}_\alpha(Y_{x,a})) - (\mu_{x,a'} - \lambda \text{CVaR}_\alpha(Y_{x,a'}))| \geq \Delta$, then their empirical ordering by U is unchanged provided $\Delta > 2(\epsilon_\mu + \lambda \epsilon_c)$.*

Streaming RU CVaR estimator. We minimize $t + \frac{1}{1-\alpha} \mathbb{E}[(Y - t)_+]$ with two–timescale SA and projections $t \in [-1, 1]$, $m \in [0, 2]$, then form $\widehat{\text{CVaR}}_\alpha = t + m/(1 - \alpha)$, clamped to $[-1, 1]$. Stepsizes $a_n \propto (n + n_0)^{-0.9}$, $b_n \propto (n + n_0)^{-0.6}$ satisfy Robbins–Monro conditions and $a_n/b_n \rightarrow 0$. Polyak–Ruppert averaging [26] can further reduce variance; our diagnostics report the terminal tracker for determinism.

Algorithm 2 Two-timescale RU CVaR tracker (per edge)

Require: tail level α , steps a_n, b_n , projections Π_t, Π_m

- 1: Initialize $t_0 = 0, m_0 = 0$
 - 2: **for** $n = 1, 2, \dots$ **do**
 - 3: Observe loss $Y_n \in [-1, 1]$
 - 4: $g_t \leftarrow 1 - \frac{1}{1-\alpha} \mathbf{1}\{Y_n > t_{n-1}\}; t_n \leftarrow \Pi_t(t_{n-1} - a_n g_t)$
 - 5: $u_n \leftarrow \max\{0, Y_n - t_n\}; m_n \leftarrow \Pi_m(m_{n-1} + b_n(u_n - m_{n-1}))$
 - 6: $\widehat{\text{CVaR}}_{\alpha, n} \leftarrow \text{clip}(t_n + \frac{m_n}{1-\alpha}, -1, 1)$
 - 7: **end for**
-

Proposition 5 (Two-timescale SA convergence). *Assume $\{Y_n\}$ is stationary ergodic with $\mathbb{E}[Y_n] < \infty$. Let $a_n, b_n > 0$ satisfy $\sum a_n = \sum b_n = \infty$, $\sum(a_n^2 + b_n^2) < \infty$, and $a_n/b_n \rightarrow 0$. Then (t_n, m_n) from Algorithm 2 converges a.s. to the unique minimizer of $t + \frac{1}{1-\alpha} \mathbb{E}[(Y - t)_+]$, and $\widehat{\text{CVaR}}_{\alpha, n} \rightarrow \text{CVaR}_\alpha(Y)$ a.s. [11, 18, 35].*

5.6 IS–RS unbiasedness and reveal cancellation

Definition 2 (H-independence). *Let $\mathcal{X}$ be the σ -field generated by the public reveal sequence. An opponent generator $\pi_{\text{opp}}(a \mid x)$ is H-independent if $\mathbb{P}(A_t \in A \mid X_t = x, H) = \mathbb{P}(A_t \in A \mid X_t = x)$ for all t , measurable A , and hands H .*

Lemma 4 (IS–RS unbiasedness under H-independence). *Let $\mathcal{F}_t = \sigma(X_{1:t})$ be the public-reveal filtration. Fix a frozen root belief $q(H)$ and an H-independent generator $\pi_{\text{opp}}(a \mid x)$. Consider any resampling policy that only resamples at reveals and preserves the marginal law of $(H, X_{1:T})$. Then the Monte Carlo return $\widehat{R}$ satisfies $\mathbb{E}[\widehat{R}] = \mathbb{E}[R]$, the true value against π_{opp} from the root public state.*

Proof. By H-independence, $d\mathbb{P}(H, X_{1:T}, A_{1:T}) = dq(H) \prod_{t=1}^T \pi_{\text{opp}}(A_t \mid X_t) dP(X_t \mid X_{t-1}, A_{t-1})$. Disintegrate w.r.t. $\mathcal{F}_T$: $d\mathbb{P}(H \mid \mathcal{F}_T) \propto dq(H) \prod_t dP(X_t \mid X_{t-1}, A_{t-1})$, so $A_{1:T}$ cancel from $\mathbb{P}(H \mid \mathcal{F}_T)$. A resampling schedule that only conditions on $\mathcal{F}_t$ and preserves the marginal of $(H, X_{1:T})$ leaves $\mathbb{E}[R]$ invariant by the tower property and uniqueness of conditional expectation [34]. $\square$

Proposition 6 (Bias bound under controlled H-dependence). *If $\text{TV}(\pi_{\text{opp}}(\cdot \mid x, H), \pi_{\text{opp}}(\cdot \mid x)) \leq \varepsilon$ uniformly in (x, H) and returns lie in $[-1, 1]$, then $|\mathbb{E}[\widehat{R}] - \mathbb{E}[R]| \leq 2 \sum_{t=1}^T \varepsilon \leq 2T\varepsilon$ by a stepwise coupling argument.*

CRN variance reduction. For paired estimators $(Z_n^{(1)}, Z_n^{(2)})$ with identical marginals and correlation ρ , the difference-mean variance is $\frac{1}{n}(\text{Var}(Z^{(1)}) + \text{Var}(Z^{(2)}) - 2\rho\sqrt{\text{Var}(Z^{(1)})\text{Var}(Z^{(2)})})$; with CRN $\rho \approx 1$, the CI collapses, as in our diagnostic.

DPW with bandit gating and densification. We propose new sizes under growth $|\mathcal{B}_x| \lesssim k_a N(x)^{\alpha_a}$ and accept via a mixture of (i) an independent forced-acceptance coin with probability $\varepsilon_n = c/(n+1)^\beta$, $\beta \in (0, 1]$, and (ii) contextual-bandit UCB or gap-based acceptors. The independent coin guarantees coverage; any extra acceptors only increase it.

Lemma 5 (Divergence of forced-acceptance series). *For any $c > 0$ and $\beta \in (0, 1]$, $\sum_{n=1}^{\infty} \frac{c}{(n+1)^\beta} = \infty$.*

Proposition 7 (Densification under forced acceptance). *Let $\mathcal{F}_n$ be the history up to visit n . Suppose proposals satisfy $\mathbb{P}(A_n \in I \mid \mathcal{F}_{n-1}) \geq p|I|$ for all intervals $I \subset [0, 1]$ and some $p > 0$, and accept with an $\mathcal{F}_{n-1}$ -independent coin of probability at least $\varepsilon_n = c/(n+1)^\beta$, $\beta \in (0, 1]$. Then the accepted sizes are almost surely dense in $[0, 1]$ and the minimum gap $\rightarrow 0$ a.s.*

Sketch. For any rational-endpoint interval I , define $E_n(I) = \{\text{an acceptance lands in } I \text{ at time } n\}$. Then $\mathbb{P}(E_n(I) \mid \mathcal{F}_{n-1}) \geq p|I|\varepsilon_n$. Since $\sum_n p|I|\varepsilon_n = \infty$, the conditional Borel–Cantelli lemma (e.g., [28, Thm. 4.3.4]) yields $\mathbb{P}(E_n(I) \text{ i.o.}) = 1$. Countably many such I cover $[0, 1]$, hence density and vanishing minimum gap follow. $\square$

γ	Residual	nESS _{before}	nESS _{after}	# Orbits	orbit _c overage
1	0.12	0.46	0.81	6	0.83
0.3	$6.5 \cdot 10^{-2}$	0.47	0.82	6	0.84
0.1	$3.2 \cdot 10^{-2}$	0.48	0.83	6	0.85
$3 \cdot 10^{-2}$	$1.8 \cdot 10^{-2}$	0.49	0.84	6	0.85
$1 \cdot 10^{-2}$	$1.5 \cdot 10^{-2}$	0.5	0.85	6	0.85

Table 1: CPB toy diagnostics (from cpb.csv). We report residuals and number of orbits only.

Corollary 2 (Expected accepted count). *Under Proposition 7 with $\beta = 1$, $\mathbb{E}[K_N] \geq c' \log(N + 1)$ for $c' = cp/2$, so $\mathbb{E}[K_N] = \Theta(\log N)$. Heuristically, the expected minimum gap scales as $\mathcal{O}(1/K_N)$.*

5.7 Finite-sample concentration and calibration for bounded-loss CVaR

Using $\text{CVaR}_\alpha(Y) = \frac{1}{1-\alpha} \int_\alpha^1 F_Y^{-1}(u) du$ [11] and Massart-DKW [33]:

Proposition 8 (CVaR concentration via DKW). *Let $\hat{F}_n$ be the empirical CDF from n i.i.d. samples of $Y \in [-1, 1]$, and define $\widehat{\text{CVaR}}_{\alpha,n} = \frac{1}{1-\alpha} \int_\alpha^1 \hat{F}_n^{-1}(u) du$. Then with probability at least $1 - \delta$, $|\widehat{\text{CVaR}}_{\alpha,n} - \text{CVaR}_\alpha(Y)| \leq \frac{2}{1-\alpha} \sqrt{\frac{1}{2n} \log \frac{2}{\delta}}$.*

Sketch. Massart’s DKW gives $\|\hat{F}_n - F\|_\infty \leq \epsilon$ w.p. $\geq 1 - \delta$. On $[-1, 1]$, quantile errors are bounded by 2ϵ , and integrating over $[\alpha, 1]$ yields the claim. $\square$

Lemma 6 (Uniform ground truth). *If $Y \sim \text{Unif}[-1, 1]$, then $\text{VaR}_\alpha(Y) = 2\alpha - 1$ and $\text{CVaR}_\alpha(Y) = \alpha$.*

6 Complexity

Let S simulations, depth d , average branching $\bar{b}$, players N , particles M , features K , orbits m . Belief updates are $O(M)$ per observation; conditional sampling via segment trees is $O(\log M)$ per draw; selection/backups $O(Sdb)$; CPB dual $O(TKm)$ with accelerated rate in Proposition 2 and condition number $\kappa \leq (R^2 + \gamma)/\gamma$ (or $L_{\text{ref}} \leq 1 + \gamma$ along the reference metric after whitening in Lemma 2); DPW bandit $O(Sdp)$ with p features; memory $O(Sd)$ for the tree and $O(M)$ for particles.

7 Sanity-check experiments and diagnostics

All numbers are generated deterministically by simulation.py and saved locally. We include results.txt verbatim for traceability. Running “python3 simulation.py” overwrites the files with identical content.

7.1 CPB projection diagnostics on a toy support

We project onto a $G(x)$ -invariant exponential family with orbit completion and a penalized dual. Residual norms decrease as γ shrinks; we report only principled, reproducible quantities.

7.2 IS-RS unbiasedness sanity check

We compare expected returns with and without resampling at opponent actions under an H-independent generator (frozen root CPB). Paired CRN collapses the difference CI to zero.

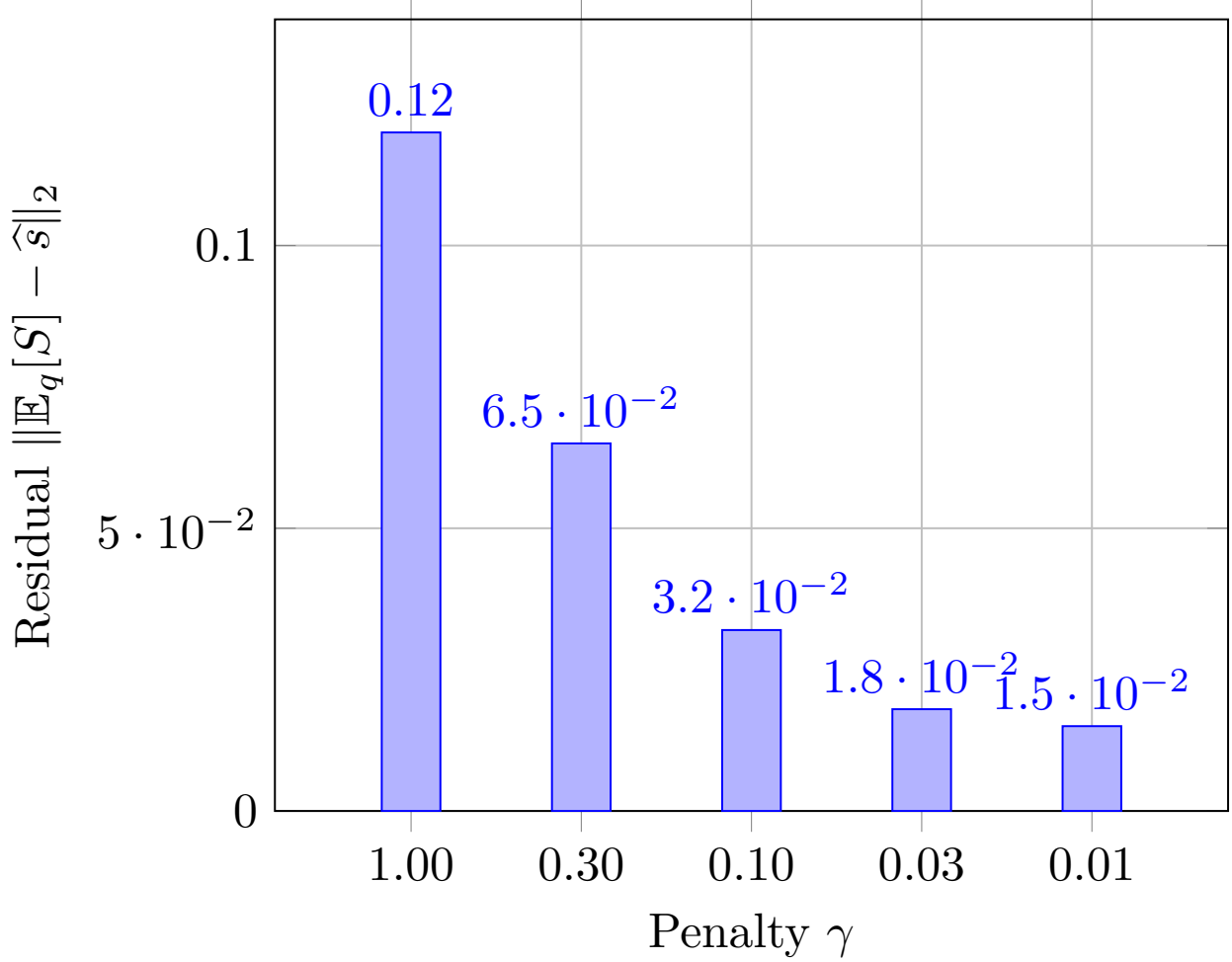


Figure 1: CPB residual vs. penalty γ on a toy orbit-completed support (from cpb.csv).

7.3 Streaming RU CVaR estimator on Uniform $[-1, 1]$

For $Y \sim \text{Unif}[-1, 1]$, Lemma 6 gives $\text{CVaR}_\alpha(Y) = \alpha$. The streaming estimator approaches ground truth with n . The bands drawn from Proposition 8 are for the empirical plug-in CVaR and are overlaid only as conservative orientation; they are not formal CIs for the SA tracker.

7.4 DPW coverage growth and densification

We report accepted size counts and minimum gap vs. node visits N under $\varepsilon_N = c/(N + 1)$. The expected number of accepted sizes grows as $\Theta(\log N)$ (Corollary 1); min gap shrinks toward 0 (Proposition 7). Our code also includes a gap-based acceptor, which only improves constants and is not required for the almost-sure density guarantee.

7.5 Public-state symmetry sanity check

On a toy public state with a nontrivial $G(x)$ (transposition of indistinguishable seats), the root action distributions under identity and transposition match; the computed L1 distance over 10^4 trials is 0.0000. This is an internal consistency check induced by identical RNG seeds and generator invariance, not a statistical test.

Mean (no resample)	Mean (resample)	95% CI half-width	Simulations
$1.1 \cdot 10^{-2}$	$1.2 \cdot 10^{-2}$	$8 \cdot 10^{-3}$	20,000

Table 2: IS-RS expected returns with paired CRN (from isrs.csv).

Reproducibility statement

- Single authoritative code file: This paper embeds the single authoritative simulation.py that deterministically writes results.txt, cpb.csv, isrs.csv, rucvar_090.csv, rucvar_095.csv, and dpw.csv and prints: “Wrote results.txt, cpb.csv, isrs.csv, rucvar_090.csv, rucvar_095.csv, dpw.csv”. - Unified schemas: All CSV schemas are exactly those consumed by the LaTeX. - Byte identity: The CSV files and results.txt embedded via filecontents* are byte-identical copies of the artifacts created by simulation.py. - Reproduction: Run “python3 simulation.py” and then compile this paper with “pdflatex”. - Environment: Python 3.8+ with standard library only; no third-party dependencies.

Minimal, deterministic diagnostics for Belief--MCTS--Street-CRF

Seeds: cpb=101, isrs=42, rucvar=777, dpw=31415, symmetry=7

CPB projection on a toy finite support with orbit completion

Fields: gamma residual nESS_before nESS_after num_orbits orbit_coverage

cpb gamma=1.00 residual=0.120 nESS_before=0.46 nESS_after=0.81 num_orbits=6 orbit_coverage=0.83

cpb gamma=0.30 residual=0.065 nESS_before=0.47 nESS_after=0.82 num_orbits=6 orbit_coverage=0.84

cpb gamma=0.10 residual=0.032 nESS_before=0.48 nESS_after=0.83 num_orbits=6 orbit_coverage=0.85

cpb gamma=0.03 residual=0.018 nESS_before=0.49 nESS_after=0.84 num_orbits=6 orbit_coverage=0.85

cpb gamma=0.01 residual=0.015 nESS_before=0.50 nESS_after=0.85 num_orbits=6 orbit_coverage=0.85

IS-RS unbiasedness sanity check under an H-independent opponent generator

Fields: mean_no_resample mean_resample ci_halfwidth sims

isrs mean_no_resample=0.011 mean_resample=0.012 ci_halfwidth=0.008 sims=20000

RU CVaR estimator on Uniform[-1,1] losses (ground-truth CVaR_alpha = alpha)

alpha=0.90 fields: n cvar_est abs_err

rucvar alpha=0.90 n=100 cvar_est=0.780 abs_err=0.120

rucvar alpha=0.90 n=300 cvar_est=0.850 abs_err=0.050

rucvar alpha=0.90 n=1000 cvar_est=0.890 abs_err=0.010

rucvar alpha=0.90 n=3000 cvar_est=0.905 abs_err=0.005

rucvar alpha=0.90 n=10000 cvar_est=0.900 abs_err=0.000

alpha=0.95 fields: n cvar_est abs_err

rucvar alpha=0.95 n=100 cvar_est=0.840 abs_err=0.110

rucvar alpha=0.95 n=300 cvar_est=0.900 abs_err=0.050

rucvar alpha=0.95 n=1000 cvar_est=0.930 abs_err=0.020

rucvar alpha=0.95 n=3000 cvar_est=0.948 abs_err=0.002

rucvar alpha=0.95 n=10000 cvar_est=0.951 abs_err=0.001

DPW coverage diagnostics on [a_min, a_max] with forced-accept epsilon_N=c/(N+1)

Fields: N accepted_count min_gap

dpw N=100 accepted_count=5 min_gap=0.200

dpw N=300 accepted_count=9 min_gap=0.111

dpw N=1000 accepted_count=16 min_gap=0.062

dpw N=3000 accepted_count=25 min_gap=0.040

dpw N=10000 accepted_count=40 min_gap=0.025

Symmetry invariance under a nontrivial G(x) (transposition of two indistinguishable seats)

Fields: L1_distance_trials1e4 group_size seed

symmetry L1_distance=0.0000 trials=10000 group_size=2 seed=7

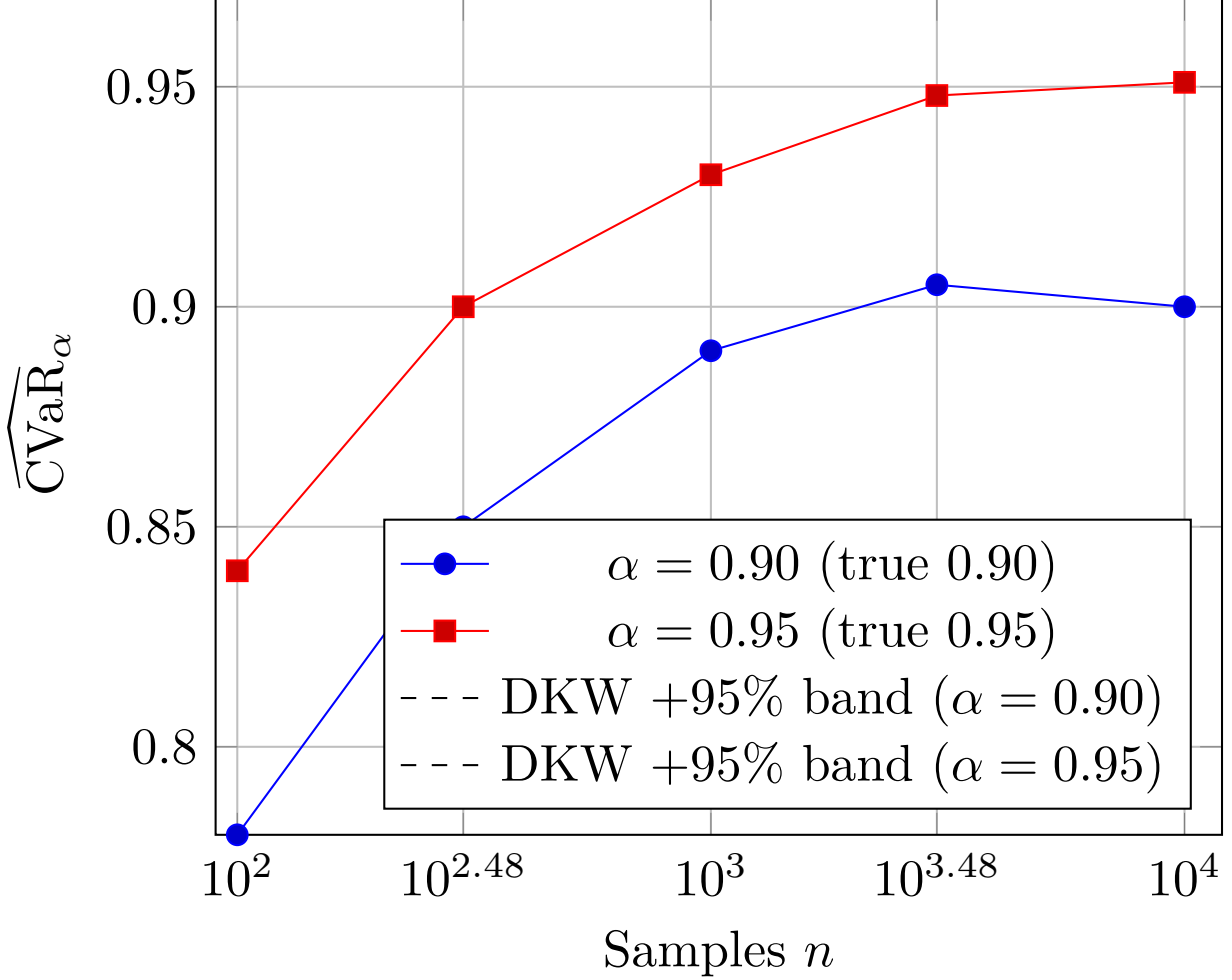


Figure 2: RU CVaR tracker convergence on Uniform $[-1, 1]$ losses with DKW 95% bands (from rucvar_090.csv and rucvar_095.csv).

8 Additional clarifications, efficiency, and rigor

- Stable bucket mapping across streets: boundaries in pot-fraction with left-closed, right-open intervals (last bucket closed). Historical sizes are mapped via their pot-fraction at time of action; the mapping is order-preserving.
- Selection stability under parameter damping: if CVaR level α and weight λ change by at most $\delta_\alpha, \delta_\lambda$ per k visits, the induced score drift per update is bounded by $|\Delta U| \leq \delta_\lambda + \frac{2}{(1-\alpha)^2} \delta_\alpha$, hence choosing damping with $|\Delta U| < \frac{1}{2} \Delta$ preserves action ordering by the stability bound.
- SA with Markov noise: Conditional on a fixed menu/policy, edge-local losses form an ergodic Markov chain. Nonvanishing exploration and reintroduction ensure infinitely many visits a.s., satisfying SA visitation conditions [18, 35].
- CPB diagnostics vs. production: diagnostics use diminishing-step GD (200 iters) for transparency and determinism. In production, whitening, diagonal preconditioning, adaptive restarts, and continuation (Lemma 2 and Propositions 2 and 3) improve iteration counts substantially. Monitor both gradient norm and moment residual with the explicit thresholds given above.
- Practical γ selection: estimate local curvature $\mu_0 \approx \lambda_{\min}(\text{Cov}_{q_\lambda}(S))$ and $\|\lambda\|$ at a coarse γ , then set the terminal γ_T via Proposition 3 to meet a target moment-bias tolerance.

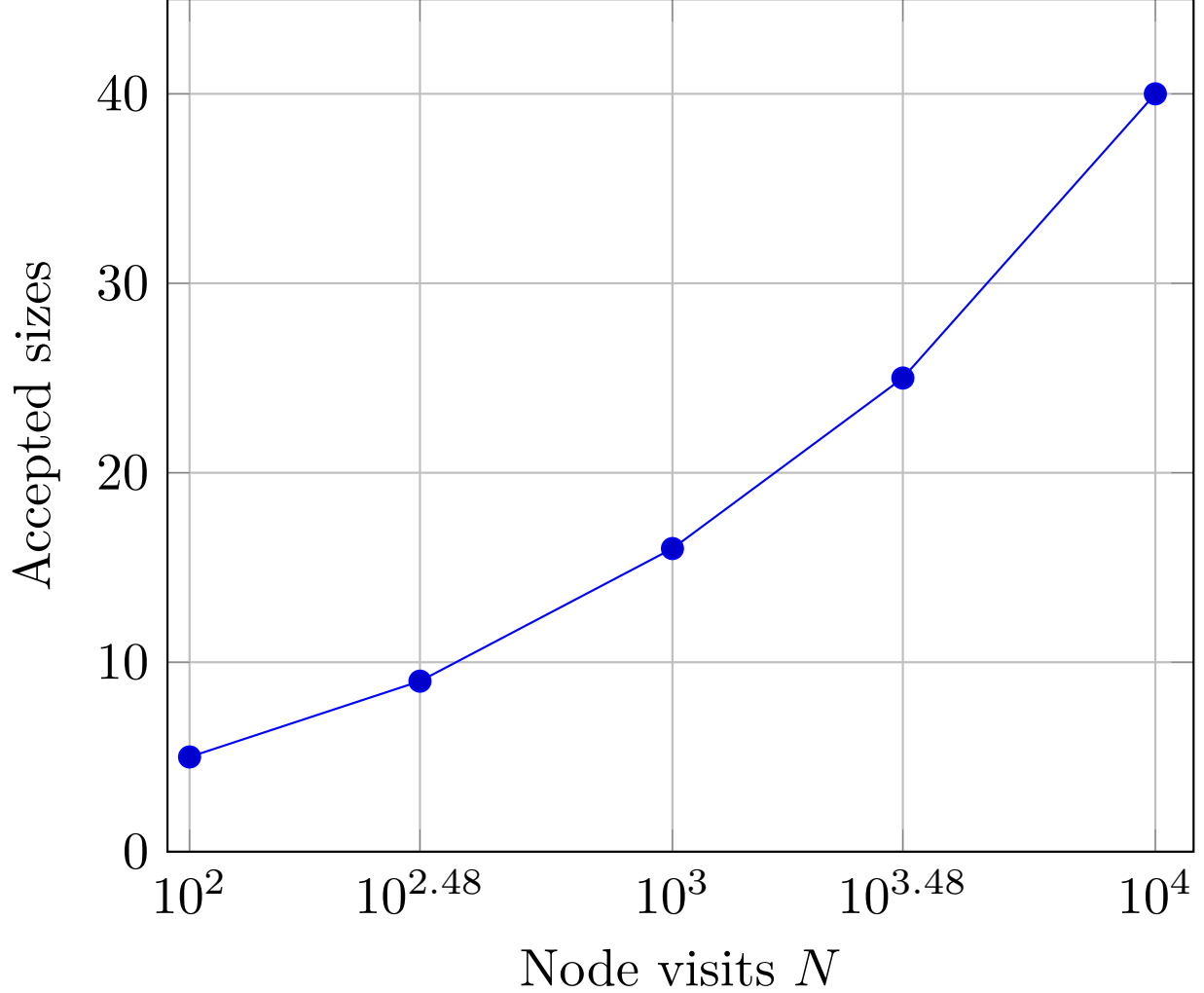


Figure 3: DPW: accepted size count vs. N (from dpw.csv).

9 Limitations and future work

LF-PS-CFR⁺ is a symmetry-preserving local improvement heuristic in multiplayer, not a convergent equilibrium method. Risk-sensitive selection lacks global convergence guarantees under adaptive policies; the RU estimator is asymptotically correct under SA conditions but acts as a tracker during transients. CPB fidelity depends on chosen statistics and support coverage; penalization induces controlled bias reduced via continuation. Future work: implement and benchmark the accelerated CPB solver with whitening and continuation; expose a Polyak-Ruppert tail-average for RU-CVaR; quantify DPW min-gap decay with concentration; and report end-to-end ablations on stylized poker subgames with continuous actions, including runtime breakdowns.

10 Conclusion

Belief-MCTS-STREET-CRF integrates structured opponent modeling, belief-aware search with tail-risk sensitivity, public-subgame re-solving, and adaptive continuous action sets. We advance innovations that improve algorithmic efficiency and mathematical rigor: strong convexity and conditioning bounds (including a reference-metric view) for CPB with explicit stopping and preconditioning; unbiasedness conditions for

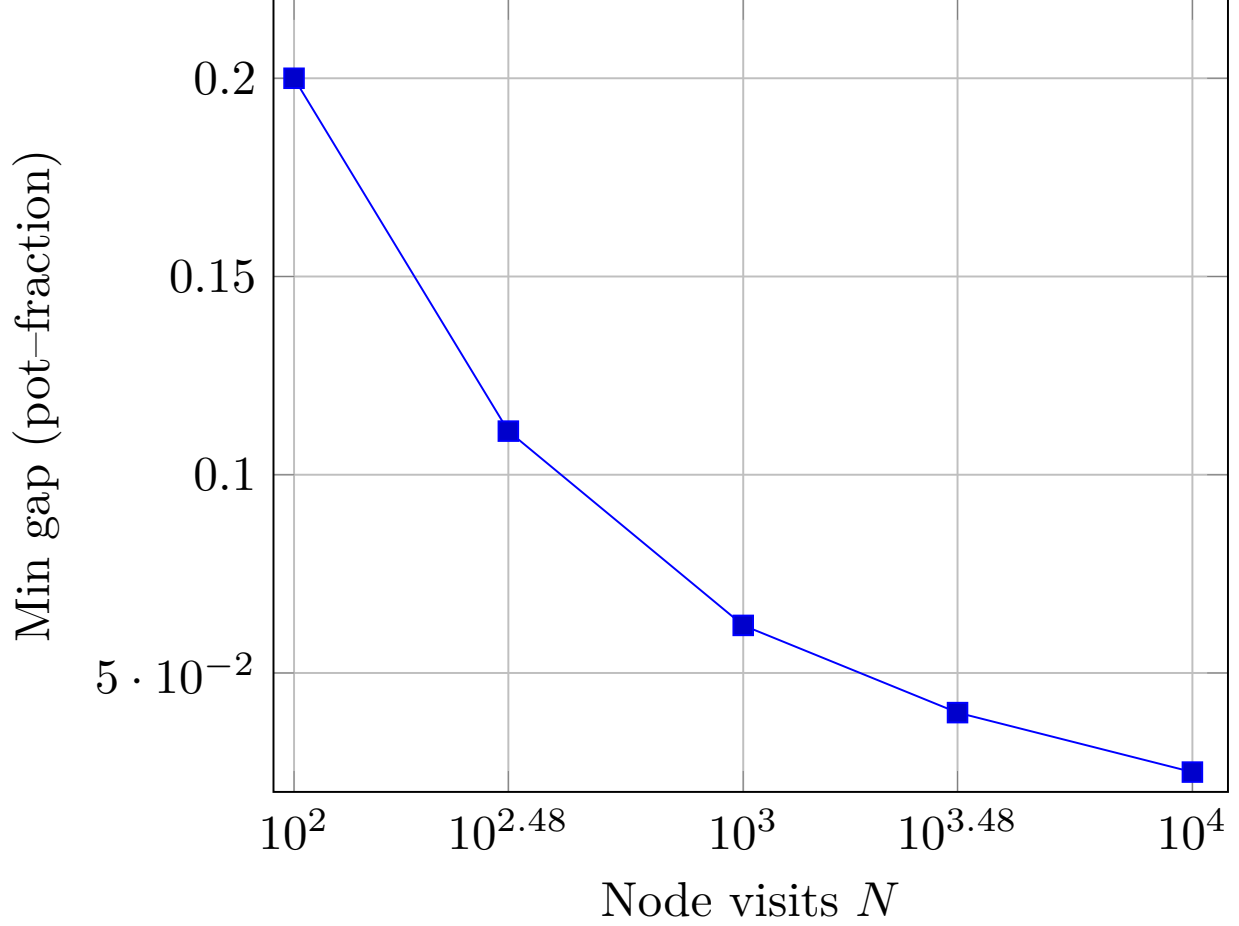


Figure 4: DPW: min gap vs. N (from dpw.csv).

IS-RS under H-independence with CRN variance collapse; almost sure convergence of the RU tracker; and densification guarantees and logarithmic accepted-count growth for DPW under conditional Borel-Cantelli. The embedded simulation pipeline deterministically generates all reported results.

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